

15. CONCEPT DEMO

SGBA Visual User Flow and Screen Scenarios

One UI · Stealth Gesture-Based Access (SGBA) | Registration No: 2350329077

15.1 Section Purpose

This section visualizes step by step the real user interaction scenario of the SGBA system. It has been prepared to concretely convey the system's operation and user experience without a physical prototype.

Core Principle — An outside observer cannot detect that the user is operating any security layer. The system runs entirely invisibly.

15.2 User Interaction Flow

The table below illustrates every step from a user activating the SGBA system to accessing the Hidden Panel.

#	Step	Description
1	Home Screen Active	The user picks up the unlocked device. The normal home screen is displayed — no security icon, folder, or badge is present.
2	Gesture Initiated	The user places a finger on an empty area of the screen. One UI Home begins tracking X/Y coordinates in the background via a low-level event listener.
3	Vector Recording	When the finger movement is complete, the system converts the gesture into a vector array. No line or trace appears on screen.
4	DTW Comparison	The Dynamic Time Warping algorithm compares the input vector against the stored reference vector. The match threshold is 85% by default.
5	Result	Match successful → Hidden Panel opens. Match failed → the system remains unresponsive. An external observer cannot determine what occurred.

15.3 Screen Flow — 3-Step Concept

The three screens below summarize the SGBA experience from the user's perspective:

Screen	Title	Description
Screen 1	Normal Home Screen	Standard app grid is visible. No Secure Folder icon, no SGBA badge. Completely innocuous appearance from the outside.
Screen 2	Settings → Invisible Access	Settings > Security & Privacy > Invisible Access menu. Gesture recording, tolerance adjustment, and Knox biometric options are configured here.
Screen 3	Hidden Panel Opened	After the correct gesture is completed, the Hidden Panel appears. Applications concealed within the Knox-secured area become accessible.

15.4 Gesture Matching Mechanism — DTW

Dynamic Time Warping (DTW), the core algorithm of SGBA, flexibly compares two different time series. This allows the user's own gesture to be recognized regardless of variations in speed or pressure.

Feature	Standard SGBA	SGBA + AI Layer
Threshold Setting	Fixed (85%)	Dynamic & Personalized
Unknown User Detection	No	Yes (Adaptive Mode)
Processing Unit	CPU	NPU (Low Power)
Battery Usage	< 2% CPU	< 1% (NPU Optimized)

15.5 Concept Demo — Security Summary

The concept demo covers not only technical security mechanisms but also the social security dimension:

Zero Visual Trace — No line, effect, or feedback appears on screen during gesture drawing. An external observer cannot become aware of the system's existence.

Non-Responsiveness on Failed Attempts — When an incorrect gesture is entered, the system gives no response — no error message, no vibration. This behavior renders brute-force attacks ineffective.

Knox Dual Shield Integration — When the optional Knox biometric layer (fingerprint / face recognition) is activated, the system offers an 'Invisibility + Biometric Security' dual shield.